Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Quantify the power loss of a median split on a continuous predictor.
- Demonstrate regression to the mean in a simulated pre-post design.
- Distinguish apparent improvement due to RTM from a true treatment effect.

Prerequisites

Basic regression; ANCOVA.

Background

Dichotomising a continuous predictor — *high vs low* — loses information in proportion to the variability thrown away. A well-known rule of thumb is that the median split reduces power in a comparison equivalent to throwing away roughly a third of the sample. The only time a dichotomy is defensible is when the clinical decision it feeds into is itself binary, and even then the continuous underlying variable should enter the analysis.

Regression to the mean is a statistical fact about correlated pairs of measurements: participants who score unusually high on a first measurement will, on average, score less extremely on a second measurement of the same quantity, even with no intervention. In a pre-post study this produces apparent *improvement* in the high baseline group and apparent *worsening* in the low baseline group. ANCOVA handles this automatically; a simple comparison of pre and post in the extreme group does not.

RTM is not the same as measurement error, but measurement error is one source of RTM. Any non-perfect correlation between measurements produces it. This is why baseline stratification by a continuous variable must be prespecified — selecting on a single noisy baseline guarantees RTM.

Setup

```r
library(tidyverse)
library(broom)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Two simulations:

1. A continuous predictor truly associated with a continuous outcome; compare the *p*-value when treated as continuous vs median-split.
2. A pre-post dataset with no intervention; show apparent improvement in the high-baseline group.

2. Visualise

```r
x <- rnorm(n)
y <- 0.3 * x + rnorm(n, 0, 1)
dat1 <- tibble(x, y, x_bin = factor(ifelse(x > median(x), "high", "low")))

ggplot(dat1, aes(x, y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", colour = "steelblue") +
  labs(x = "x (continuous)", y = "y")
```

3. Assumptions

Independence, approximate normality; standard regression assumptions.

4. Conduct

Power loss from dichotomisation:

```
tidy(lm(y ~ x_bin, data = dat1)) # median split
```

The p-value for x_bin should be larger than for x; the standard error is larger and the estimate is attenuated.

Regression to the mean:

```
pre  <- rnorm(n2, 100, 15)
post <- 0.7 * (pre - 100) + 100 + rnorm(n2, 0, 11)
dat2 <- tibble(pre, post) |>
  mutate(high_baseline = pre > quantile(pre, 0.8))

dat2 |>
  group_by(high_baseline) |>
  summarise(pre_mean = mean(pre), post_mean = mean(post),
            change = mean(post - pre))
```

The “high baseline” group shows a large negative mean change; the rest of the sample shows a small positive change. No intervention occurred.

```
  geom_point(alpha = 0.3) +
  geom_abline(slope = 1, intercept = 0, linetype = 2, colour = "grey50") +
  geom_smooth(method = "lm", colour = "firebrick") +
  labs(x = "Baseline", y = "Follow-up")
```

The fitted line is flatter than the identity line; that flattening is RTM.

5. Concluding statement

> Dichotomising the continuous predictor inflated its p-value from `signif(tidy(lm(y ~ x, data = dat1))$p.value[2], 2)` to `signif(tidy(lm(y ~ x_bin, data = dat1))$p.value[2], 2)` and widened its standard error. In the pre-post simulation with no intervention, the top-baseline quintile showed a mean decline of `round(dat2 |> filter(high_baseline) |> summarise(mean(post - pre)) |> pull(), 1)` units, entirely attributable to regression to the mean.

Common pitfalls

- Dichotomising to avoid thinking about the slope.
- Comparing pre-post within a high-baseline subgroup and calling the decline an effect.
- Using a paired t-test on change in a selected subgroup.

Further reading

- MacCallum RC et al. (2002), *On the practice of dichotomization…*
- Altman DG, Royston P (2006), *The cost of dichotomising continuous…*
- Senn S (2011), *Francis Galton and regression to the mean.*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)